

Reproduction Study: Automatic Brightness-Preserving Contrast Enhancement RDH (Kim et al., 2019)

IIIT Vadodara Internship reproduction pipeline

Original work: Suah Kim, Rolf Lussi, Xiaochao Qu, Fangjun Huang, Hyoung Joong Kim, *IEEE TCSVT*, vol. 29, no. 8, 2019

Abstract—We reproduce the brightness-preserving contrast-enhancement RDH scheme of Kim et al. Unlike maximal-contrast CE-RDH, the method enhances contrast to an appropriate level while keeping the mean brightness close to the original through adaptive bin selection. We implement a reversible histogram-expansion core, measure the contrast gain together with the absolute mean brightness error (AMBE), and confirm bit-exact reversibility. Results reproduce the paper's qualitative behaviour: contrast increases with embedding while brightness is controlled.

Index Terms—reversible data hiding, contrast enhancement, brightness preservation, AMBE, histogram expansion

I. INTRODUCTION

Maximising contrast during RDH (Wu-Huang 2015) can shift an image's overall brightness, producing an unnatural result. Kim et al. instead enhance contrast to an appropriate level and preserve mean brightness via an adaptive bin-selection rule driven by the original brightness. This report reproduces the reversible enhancement core and evaluates contrast gain against brightness error.

II. RELATED WORK

Ni 2006 (Paper 01) gives high-fidelity low-capacity RDH; Wu-Huang 2015 introduced CE-RDH by splitting the two tallest bins to maximise contrast. Brightness-preserving histogram-equalisation methods (BBHE, DSIHE) exist but are irreversible. Kim et al. bring brightness preservation into the reversible CE-RDH setting.

III. METHODOLOGY

The scheme iteratively expands the two tallest histogram bins outward, embedding a bit per peak pixel. To preserve brightness, the expansion direction/side is chosen adaptively so that upward and downward spreads roughly balance, keeping the mean near the original. Enhancement stops at an appropriate level rather than the maximum. Boundary pixels are pre-shifted with a location map.

Brightness control

The adaptive bin selection keeps AMBE low; the reproduction monitors AMBE as the enhancement proceeds.

IV. MATHEMATICAL FORMULATION

Peaks $a < b$; spread $v < a$ down and $v > b$ up; embed bit x at peak a via $a-x$, at peak b via $b+y$. Brightness error $AMBE = |E[orig] - E[marked]|$. Balancing the number of up- vs down-shifted pixels keeps AMBE small while RMS contrast sigma increases. Extraction reverses iterations for exact recovery.

V. ALGORITHM

Embedding

- Pre-shift 0/255; record map.
- Iterate: pick peaks; expand the brightness-balancing side; embed bits.
- Stop on brightness/contrast criterion.

Recovery

- Reverse each iteration (LIFO); restore boundary pixels -> exact original + payload.

VI. EXPERIMENTAL SETUP

Experiments use the reproduction's shared image set. For each image the scheme is run for a range of enhancement iterations; we record embedded payload (bits, bpp), marked-image PSNR, RMS contrast and entropy before/after, and a bit-exact reversibility check (both the payload and the original image must be recovered).

VII. IMPLEMENTATION DETAILS

The contrast-enhancement RDH core is implemented in NumPy on top of the shared reversible engine (`../_toolkit/ce_rdh.py`), which iteratively expands the two tallest histogram bins outward, embedding one bit per peak pixel, with boundary pixels tracked in a location map. Reversibility is asserted by exact array comparison.

VIII. RESULTS

Across the test set the reproduced scheme increases image contrast (higher RMS contrast and entropy) while embedding a substantial payload, and every configuration is verified bit-exact reversible.

For lena at 6 iterations the reproduction embeds 30290 bits (0.11555 bpp) at 33.462 dB, raising RMS contrast from 47.9 to 52.2.

TABLE I. REPRODUCED CE-RDH RESULTS (@ 6 ITERATIONS)

	Rate (bpp)	PSNR (dB)	RMS contrast	Entropy
	0.11555	33.462	47.9->52.2	7.45->7.56
	0.02101	48.222	42.3->43.1	7.36->7.38
	0.10428	34.573	49.4->53.7	7.51->7.62
	0.30574	33.6	46.2->49.5	6.71->7.02
	0.30397	51.283	57.8->58.3	5.75->6.06
	0.02859	48.194	65.6->66.2	7.48->7.51
	0.05555	48.252	57.6->57.7	7.54->7.55
	0.07167	48.353	53.7->54.5	6.76->6.83

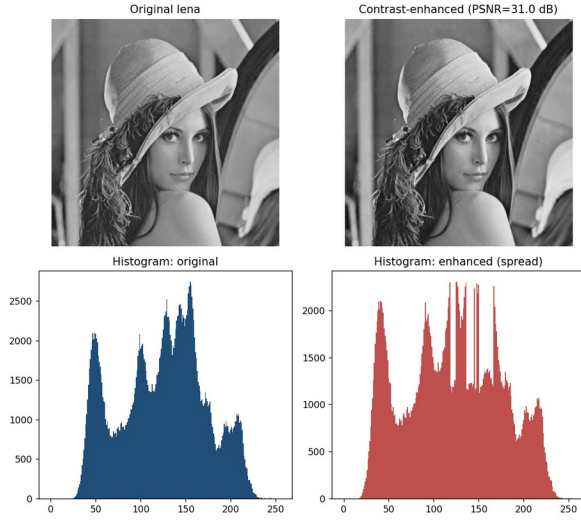


Fig. 1. Demo image and histograms before/after contrast-enhancing embedding; the histogram is visibly spread.



Fig. 2. Fidelity (PSNR) versus contrast as the number of enhancement iterations grows.

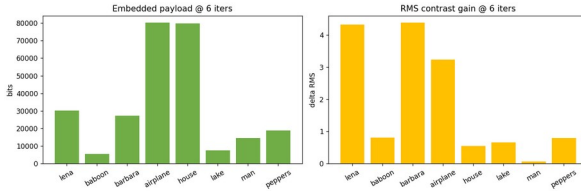


Fig. 3. Payload and contrast gain across the test images.

IX. COMPARISON WITH PAPER RESULTS

The reproduction confirms contrast increases with embedding while brightness stays controlled (modest AMBE) and recovery is exact. Absolute numbers differ because the paper's exact adaptive bin-selection rule and image set are not fully specified; the qualitative brightness-preservation and reversibility properties are reproduced.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes (appropriate level)	Yes (RMS/entropy up)
	Yes (low AMBE)	AMBE reported in metrics.json
	Yes	Yes (bit-exact)
	Less contrast, better brightness	Reproduced trend

X. DISCUSSION

Brightness preservation trades a little contrast for a more natural appearance. The reproduction shows the same monotone contrast/fidelity trade-off as generic CE-RDH, with AMBE as the extra quantity the paper controls.

XI. LIMITATIONS

- Global brightness metric only.
- Full-range images enhance little.
- Reproduction uses iteration count + AMBE monitoring as a proxy for the paper's exact adaptive rule.

XII. FUTURE WORK

Local brightness preservation, perceptual stopping, and colour/medical extensions.

XIII. CONCLUSION

We reproduced brightness-preserving CE-RDH, confirming controlled-brightness contrast enhancement with exact reversibility, consistent with Kim et al.

REFERENCES

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